Big Data Tools & Management

Prediction of Beer Trendlines and Style Popularity over 1996 to 2018

<u>Group 14</u>

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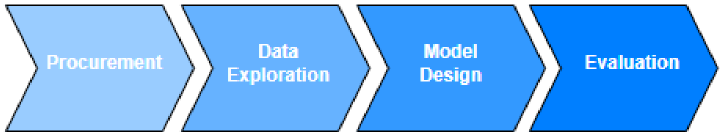
1. Background and Objectives

The history of alcohol, particularly beer, spans back tens of thousands of years. Beer production is widely dependent on the type of grain used, the yeast strain, the water, and the flavorings, which differ by the country. With innovations in modern fermentation technologies and transportation logistics, the general population can now discover beer styles from around the world from the comfort of their homes or by visiting the newest microbrewery that popped up in town.

For our project data source, we have selected a dataset from Kaggle, with a collection of 27 attributes spanned across 3 linked datasets, of beers brewed and sold worldwide during the 1996 – 2018 timeframe. The objectives of our project were to clean the dataset for processing, perform analysis using modeling, build a machine learning model to predict the popularity of a style of beer if a new brewery was to start producing, relative to its location based on data training and testing. A table containing a description for each factor is provided below:

Factor	Description
Id	Beer Identification
Name	Name of Beer
Brewery_Id	Brewery Identification
State	State
City	City
Country	Country
Availability	Year-round, Limited, Rotating, Seasonal
ABV	Alcohol By Volume
Notes	Short beer profile
Types	Brewery, Bar, Eatery, Beer-to-go
Retired	Yes/No
Username	Username of the reviewer
Date	YYYYMMDD
Text	Review of the beer
Look	Beer attribute (1 to 5)
Smell	Beer attribute (1 to 5)
Taste	Beer attribute (1 to 5)
Feel	Beer attribute (1 to 5)
Overall	Overall beer score (1 to 5)
Score	Dataset calculation

2. Process Flow Steps



In order to understand and test our findings from the dataset in a more meaningful way, the group followed the above 4 steps. Each step is summarized below:

1. **Procurement** – Dataset containing 27 aspects of beers worldwide were extracted using Kaggle. The dataset includes factors and their respective descriptions.
2. **Data Exploration** – Dataset organization, including categorizing variables, populating missing null values, and transforming the data for subsequent analysis.
3. **Model Design –** Linear Regression and Random Forest were selected as models for analysis.
4. **Model Evaluation –** Model efficiency testing by splitting dataset into two parts, 70% training and 30% testing. Performance was compared to determine the best model for testing data.

3. Data Procurement

As previously mentioned, the data source was obtained from Kaggle. Our group concluded the data size and quality to be very informative for model building, and procurement was straightforward. The dataset can be accessed here.

Upon initial examination of the dataset, it was observed that a few issues needed to be addressed before data processing could commence. The list below described some of the challenges we faced when conducting preliminary data lookups:

- Missing or null values for variables in the dataset
- One of the attributes, Score, gave no indication how values were calculated.
- Complications with reading in Databricks due to file size for Reviews

To accommodate a smoother process for data loading and reduce file size, the original Reviews .CSV file was modified with the 'Text' column removed as the information provided no significant value for our model purpose.

4. Data Exploration

Once each dataset was read into a dataframe, initial data exploration was done to gain some understanding of beer production by state and/or country. While querying the Beers dataframe, 112 unique styles were found with the most prolific type of beer to be American IPA followed by American Pale Ale, and the rarest types were Wild/Sour Beers. This information reflects what we see in the real world, with Ale-style beer being most abundant and easily available.

Using the breweries dataframe, it was found that a total of 200 countries had 45244 brewery locations, with the US dominating 66% of locations worldwide (see Figure 1).

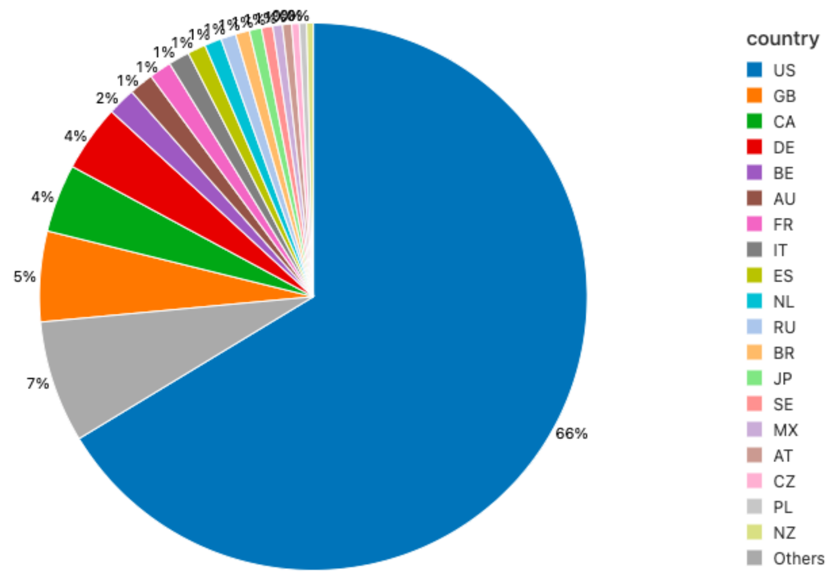


Figure 1. Pie Chart of Breweries by Country

With other countries having null or missing values for state, the group decided to delve a little deeper into breweries located in the US. Unsurprisingly, California was found to host the most per state, and North Dakoda with the least (Figure 2).

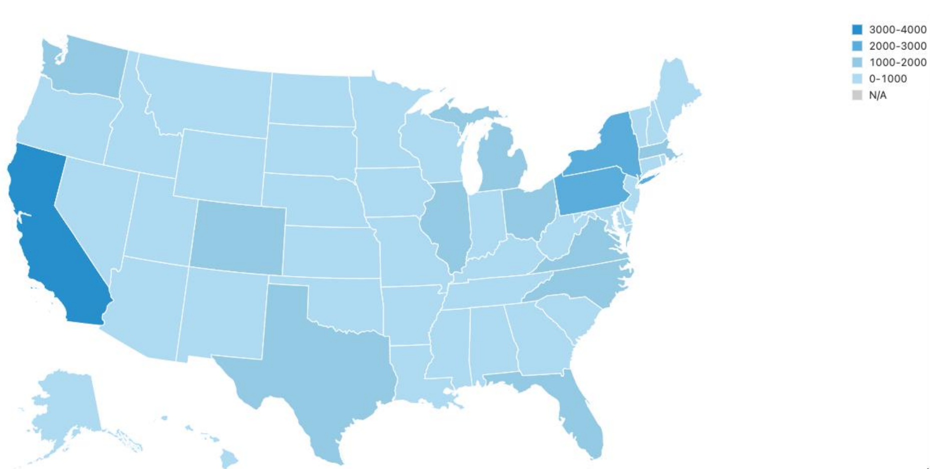


Figure 2. Number of breweries located in US by State

With both beers and breweries being primarily located in the US, it could be tentatively assumed that most reviews were also collected from US-based customers. While the Reviews dataframe did not give information regarding locations, the group took notice of a steady climb in review counts, from very sparse in the 1990s to over 1000+ daily reviews from mid-2011.

A join query was used to combine reviews, beers, and breweries dataframes by 'id' and 'brewery_id'. Extra columns generated by Databricks in the process were dropped. Incorrect values in specific fields, such as country abbreviations in Beer styles, were also removed. Records of beers with no reviews were dropped as they provided no value to the training dataset. Sanitary checks were also performed to check for 'orphan' beers with no associated breweries, and mismatches between respective State and Country columns of Beers and Breweries dataframes.

Efforts were also made to omit or replace missing values for each variable. As the data source gave no supplementary notes on how the 'Score' variable was calculated, initially the group assumed it was based on the average of the five other variables (Taste, Smell, Feel, Look, Overall). The group also noticed that Score was always populated despite the previous five variables having null values, which does not provide credibility to its values. To combat this, the group made a custom score based on the average of the previous variables, appropriately named 'Custom_Score' and it was decided to use this as the final values for model training.

Since Beer Style values were categorical variables, further transformation into numeric values was done. Finally, fields Username, Availability, Notes, Types, Retired, Look, Taste, Smell, Feel, and Overall were dropped from the final dataset in preparation for data training.

5. Model Design

To prevent data leakage in the training process, a full pipeline was built. Variables were transformed using VectorAssembler and a sample (0.5%) of the entire dataset was chosen, otherwise the split could not be processed efficiently in Databricks. Before building our model, we divided our dataset into two parts: the training set at 70% and the test set at 30%. Two models were imported for our model design, and their steps summarized below:

1. **Linear Regression Classifier**
   Training variables were standardized and the training set was run with satisfactory values for MSE, MAE, RMSE, R2 and explained variance. Holdout value was 0.13.

2. **Random Forest Classifier**
   Training set was run with satisfactory values for MSE, MAE, RMSE, R2, explained variance, but less than Linear Regression. Pipeline was added and holdout value was 0.128.

6. Model Evaluation

The evaluation of the two models chosen was done by comparing the performance of each model using the five variables below.

Linear Regression Classifier	Random Forest Classifier
MSE: 0.07700555345218366 MAE: 0.23033333049880134 RMSE Squared: 0.27749874495605137 R Squared: 0.7008922468544037 Explained Variance: 0.28826899998722283	MSE: 0.08267222001288653 MAE: 0.23577777478430006 RMSE Squared: 0.287527772594034 R Squared: 0.6788815758467652 Explained Variance: 0.27584702469159106

Looking between the two models, MSE slightly improved after increased parameters while RMSE was slightly better before pipeline application. MAE continue to be almost the same, and R Squared and explained variance values did not improve after applying the pipeline.

The holdouts were set to detect if the prediction would fall within an acceptable range from the 'Custom_Score' that was created. Results proved to be within expected limits, however holdouts did not show significant improvement after applying the pipeline which would require further data engineering not covered in this project. This means the models could be improved to be executed with more hypertuning, at the expense of longer computational time. If the group were to make a decision on which model performed better with the current results, it could be tentatively concluded that Linear Regression as it had a better R2 value.

7. Conclusion

The results from the dataset chosen helped the group to realize that this project came with more challenges than originally anticipated, due to many variables obtained after merging 3 different datasets. Despite the wealth of data contained within the combined dataset, the omission of several variables and transformations the group undertook in order to be able to make the Machine Learning process possible may have created some setbacks in data accuracy.

The two models, Linear Regression, and Random Forest with increased parameters produced satisfactory results in correlation variance but can be improved with additional hypertuning. Overall, results are as the group anticipated as it was risky to combine datasets that despite having the same topic in common, would have variables that did not correlate to each other. One improvement the group would have done early on during data exploration was to pick 1 dataset that satisfied requirements the most with the least amount of variables needed for transformation. As data types are a crucial impediment for the group to run a better model, any new data transformation executed results in potentially losing some valuable information that could be used for better results.

With more time allotted for the term project, the group would also like to investigate more model builds such as Gradient Boosted Classifiers or different hypertuning methods to improve the test dataset outcomes, and explore more platforms to analyze machine learning processes that increase efficiency while retaining adequate information.